Unit 14, Lesson 8: Making Predictions

Benchmarks

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Learning Target

☐ I can use theoretical probability to make predictions.

14.8.1 Warm-Up: Sum Probabilities

1. Find the sum of the following numbers.

$$\frac{1}{5} \qquad 0.008 \qquad 33\frac{1}{3}\% \qquad 15\% \qquad \frac{31}{500}$$

2. Explain whether or not these numbers could be the probabilities in a probability distribution.

14.8.2 Collaboration: Solving Problems with Probability

1. Three red marbles, 3 blue marbles, 3 yellow marbles, and 3 orange marbles are in a bag.

   A representation of the marbles is shown.

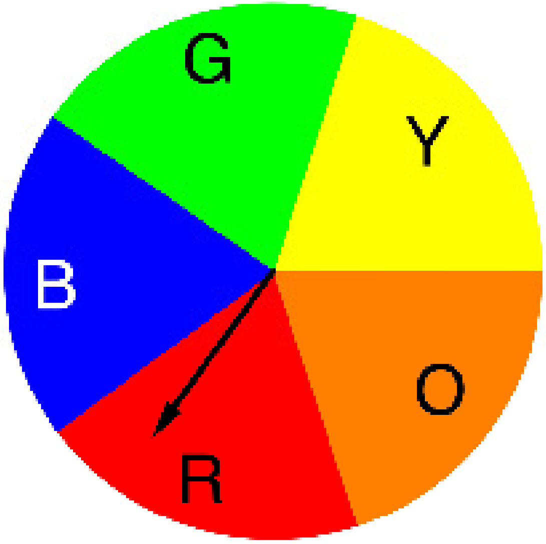
   a. If a marble is chosen 20 times, with replacement, how many times do you expect to choose a blue marble?

   b. Suppose two marbles are chosen, with replacement. What is the probability that a blue marble is chosen both times?

   c. If this sequence is repeated 40 times (for a total of 80 marbles), how many times do you expect to choose two blue marbles?

2. As part of a game, Spinner B is spun and a coin is flipped.

Spinner B



Coin

If the spinner lands on a primary color and the coin lands on heads, the player wins.

a. What is the probability of winning the game?

b. If the game is played 250 times, what is the predicted number of wins?

14.8.3 Guided Practice: Predictions in Repeated Experiments

Suppose a game is played where a fair coin is tossed 5 times.

- If none of the tosses result in heads or all of the tosses result in heads, the player wins.
- In all other outcomes, the player loses.

The probability distribution for the possible number of heads is shown below.

Number of Heads	0	1	2	3	4	5
Probability	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

1. What is the probability that a player wins on their first game?

2. What is the probability that a player loses on their first game?

3. **If a player plays this game 64 times, what is the predicted number of wins?**

4. **How many times will a player need to play the game in order to win 10 times?**

5. **If the winning prize is $20 and it costs $5 to play the game, would you want to play this game?**

Explain your reasoning.

14.8.4 Your Turn!

1. Nikita decides to buy some chicken eggs to hatch. She buys 3 eggs because she wants to have three chickens in her yard. She finds out after she orders the eggs that there is a 50% chance each egg will hatch.

   a. Find the theoretical probability that only one of the eggs will hatch.

   P(exactly one egg hatches) =

   b. Find the theoretical probability that exactly 2 of the eggs will hatch.

   P(exactly 2 eggs hatch) =

   c. How likely is it that all three of Nikita's eggs hatch?

   P(all 3 eggs hatch) =

   d. If she wants to have three chickens, explain how many eggs she should buy.

2. Marianne Douglass has a 26-card deck of cards, each with a unique letter of the alphabet on it.

   a. If she chooses two cards, with replacement, what is the probability she chooses her initials, MD, in the correct order?

   b. If she chooses a card 300 times, how many outcomes including an A are expected?

3. Ann spins a fair spinner sectioned into seven equal parts, one for each day of the week, 350 times. She correctly predicts that she will land on Monday 50 times because the theoretical probability that she lands on Monday is

$$\frac{\boxed{}}{\boxed{}}.$$

4. A fair six-sided die is rolled four times. The probability of rolling the same number on all four rolls is $\frac{1}{216}$. If this experiment is performed 1000 times (for a total of 4000 rolls), predict the number of repetitions that will result in all four rolls landing on the same number.

14.8.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.

I need help. I got it. I could help someone.

Determining the sample space for a repeated experiment.	
Finding the theoretical probability of an event related to a repeated experiment.	
Solving problems using probability.	
Making predictions based on theoretical probability.	

